

Task 4: Monitoring & Maintenance Plan

1. Monitoring Checklist

The AFL Assistant should be monitored across API performance, tool reliability, guardrails, and prediction quality.

Metric	What to Track	Alert Threshold	Cadence
Response Latency	API response time in milliseconds for each request	Warning: >2 seconds average; Critical: >5 seconds	Continuous logging; weekly review
Tool Error Rate	Percentage of retrieval/prediction tool calls that return errors	Warning: >5%; Critical: >10%	Daily/weekly
Off-topic Leak Rate	Percentage of non-AFL or prompt-injection requests that incorrectly receive an AFL answer instead of a refusal	Warning: >2%; Critical: >5%	Weekly guardrail evaluation
Prediction Accuracy	Correct match-winner predictions compared with completed match results	Investigate if accuracy falls >5 percentage points below the established validation/holdout benchmark	After each completed round
Prediction Probability Quality	Compare predicted probabilities with actual outcomes over time	Investigate sustained calibration degradation	Monthly
Data Freshness	Whether newly completed AFL matches are present in the feature/prediction tables	Alert if a completed round is missing after the weekly refresh	Weekly
API Availability	Successful vs failed API requests	Alert if availability falls below 99% during deployment	Continuous

The existing FastAPI logging records important operational fields including query, conversation ID, intent, route, tools called, latency, and errors. These logs provide the basic monitoring data required for maintenance.

2. Prediction Model Drift Monitoring

Prediction performance should be evaluated using **completed matches only**. Predictions for upcoming matches should not be evaluated until their actual results are available.

After each AFL round:

1. Add the completed match results to the AFL match dataset.
2. Recalculate the required pre-match features without using information from after the match.
3. Generate predictions for the newly completed matches using the deployed model.
4. Compare predicted outcomes with actual match winners.
5. Calculate round-level accuracy and cumulative accuracy.
6. Compare performance against the established baseline.
7. Check whether performance is consistently deteriorating.

The current Day 2 model achieved **67.13% accuracy on the 2025 holdout set**, while the same-match ladder baseline achieved **65.28%**. These values can be retained as reference benchmarks when monitoring future performance.

A single poor round should not automatically trigger retraining. Retraining should be considered when there is **sustained performance degradation**, data distribution changes, or enough new matches have accumulated to provide useful additional training information.

3. Weekly Retraining / Refresh Loop

Step 1: Ingest New Results

At the end of each completed AFL round, append the newly completed matches to the historical match dataset.



Step 2: Update Features

Recalculate rolling and historical features using information that would have been available **before each match**, including:

- Recent 5-match win rate
- Win streak
- Recent average score
- Days of rest
- Head-to-head statistics
- Pre-match ladder position

- Pre-match points
- Pre-match percentage

This prevents **data leakage** from future match results.

Step 3: Validate Data

Run automated checks for:

- Missing values
- Duplicate matches
- Invalid team names
- Incorrect dates
- Missing match winners
- Unexpected feature ranges
- Newly introduced team-name variations

If validation fails, stop the model refresh and correct the data before retraining.

Step 4: Evaluate the Existing Model

Run the current production model on the newly completed matches and record:

Round accuracy

Cumulative accuracy

Precision/recall where applicable

Prediction distribution

Baseline comparison

Step 5: Retraining Decision

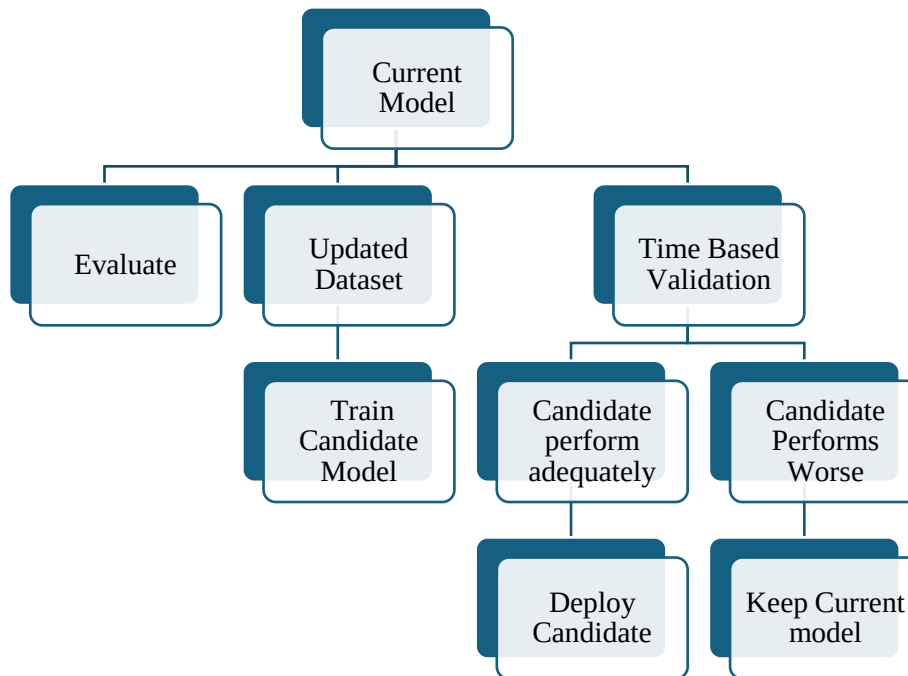
Retrain when either of the following conditions is met:

- A meaningful amount of new labeled match data has accumulated, such as a completed round or several rounds.
- Prediction performance shows sustained degradation relative to the established benchmark.
- Feature distributions have changed substantially.
- The current model is consistently underperforming the baseline.

For routine maintenance, perform a **weekly model refresh after each completed round**, while retaining the existing model if the newly trained version does not improve validation performance.

Step 6: Compare Candidate Model

Train the candidate model using the updated historical dataset and compare it with the currently deployed model on a time-based validation set.



Step 7: Version and Record

For every model update, record:

- Training-data date range
- Number of matches
- Feature version
- Model type
- Validation accuracy
- Baseline accuracy
- Training date
- Model version
- Deployment decision

This provides reproducibility and makes it possible to roll back to a previous model if necessary.

4. Maintenance Cadence

Continuous: API latency, errors, and request logs.

Daily: Review API/tool errors when the system is actively deployed.

After each AFL round: Ingest results, update features, evaluate prediction performance, and check for drift.

Weekly: Run the complete data-validation and model-refresh pipeline.

Monthly: Review cumulative prediction performance, probability calibration, guardrail evaluation, and whether the current monitoring thresholds remain appropriate.

After major system changes: Re-run the complete evaluation suite, including factual queries, prediction tests, multi-turn conversations, off-topic prompts, and prompt-injection tests.

5. Maintenance Objective

The maintenance process ensures that the AFL Assistant remains **reliable, domain-safe, data-grounded, and up to date**. New match results continuously improve the underlying dataset, while time-based validation prevents premature deployment of a model that does not improve on the existing production model.